

IN-SITU STRUCTURE OF BACTERIAL OUTER MEMBRANE PROTEIN A (OMPA) WITH FAST MAGIC-ANGLE SPINNING NMR

Tanguy LE MARCHAND¹, Johannes THOMA^{2,3}, Veronica CRESPI^{4,5} Jackson CROWLEY⁵, James TOLCHARD¹, Tobias SPARRMAN⁶, Luca MONTICELLI⁵, Björn M. BURMANN^{2,3}, Guido PINTACUDA¹

¹ Centre de Résonance Magnétique Nucléaire (CRMN) à Très Hauts Champs de Lyon (UMR 5082 - CNRS, ENS Lyon, UCB Lyon 1), 69100, Villeurbanne, France ; ² University of Gothenburg, 405 30 Göteborg, Sweden; ³ Wallenberg Centre for Molecular and Translational Medicine, 405 30 Göteborg, Sweden ; ⁴ Inserm, Pharmacology and Transplantation, UMR 1248, Centre de Biologie et Recherche en Santé, Université de Limoges, 87000, Limoges, France ; ⁵ Molecular Microbiology and Structural Biochemistry (MMSB), UMR 5086 CNRS and Université Claude Bernard Lyon 1, 7 Passage du Vercors, Lyon, France ; ⁶ Umeå University, Linnaeus väg 10, 901 87 Umeå, Sweden
(corresponding author: tanguy.le_marchand@ens-lyon.fr)

Keywords:

ABSTRACT

Outer Membranes (OM) of Gram-negative bacteria are key targets for antimicrobial research. Their investigation is however challenging for most analytical techniques due to their complex and unique structure. The outer membrane is asymmetric, composed of lipopolysaccharides (LPS) on the outer leaflet and phospholipids on the inner leaflet, which makes it difficult to mimic *in vitro*.

In this work, we investigated Outer Membrane Protein A (OmpA) expressed in OM vesicles of *E. coli* by MAS NMR. Taking advantage of a tailored isotope labelling scheme, fast MAS capabilities, and magnetic fields up to 1.2 GHz, we determined the first experimental structure of an OMP in a native OM environment. Comparison with solution NMR and X-ray diffraction structures highlights clear differences in the conformation of the extracellular loops, that can be directly linked to the more realistic membrane environment. In addition, we conducted one of the most comprehensive relaxation analyses for this class of samples in the solid state, and found significant differences in protein dynamics with respect to previous solution-NMR studies. These observations are in excellent agreement with molecular dynamics simulations

carried out in asymmetric membranes and demonstrate a clear, direct impact of the native bacterial membrane on OmpA structure and dynamics.

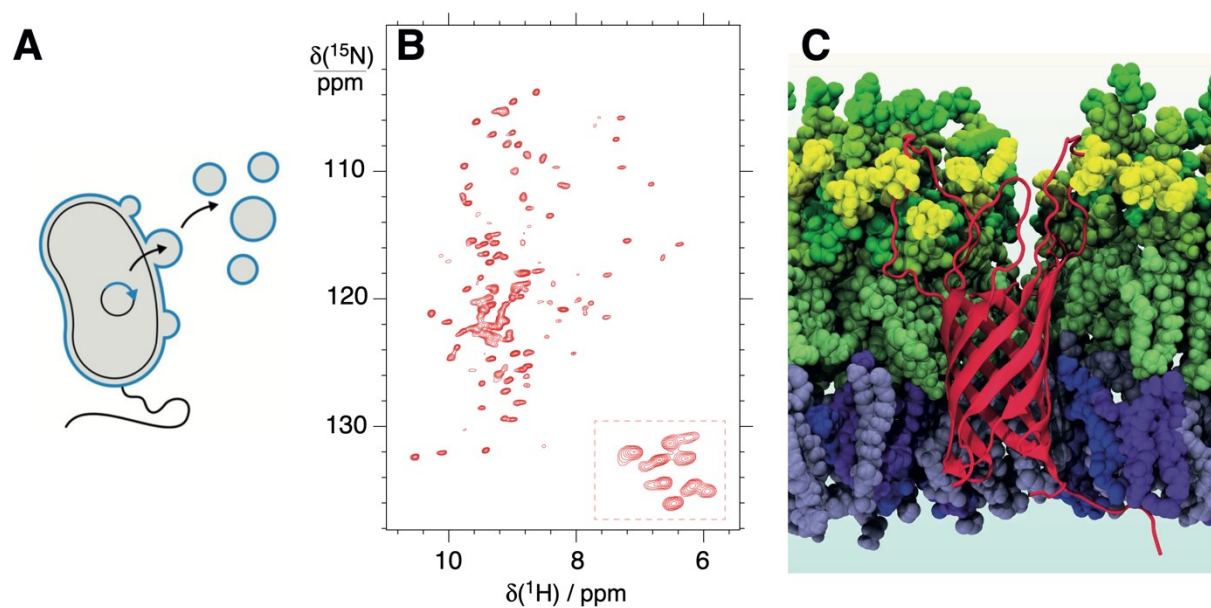


Fig. A) Schematic representation of the sample production. OmpA is over-expressed in *E. coli* outer membrane and harvested with the enriched outer membrane vesicles. B) 2D ^1H - ^{15}N correlation NMR spectrum of OmpA recorded on a 1.2 GHz spectrometer at 100 kHz magic-angle spinning. C) NMR-derived 3D structure of OmpA optimized during a molecular dynamics simulation in the outer membrane.